

# The empirical recurrence for OEIS A252389, proved by transfer matrix

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## Abstract

OEIS A252389 counts  $7 \times (n+2)$  arrays over  $\{0, \dots, 3\}$  subject to a condition imposed on every  $3 \times 3$  subblock, and it carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is not an empirical matter. A  $3 \times 3$  subblock spans three consecutive columns, so the state of a transfer matrix is a PAIR of consecutive columns and appending one more column is a step; an array is then a walk, and  $a(n)$  is a walk count on  $S = 1720956$  vertices. Such a sequence is  $C$ -finite, so whether a given recurrence annihilates it is decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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## 1 The conjecture

OEIS A252389 is “Number of  $(5+2) \times (n+2)$  0..3 arrays with every  $3 \times 3$  subblock row and diagonal sum equal to 0 3 5 6 or 7 and every  $3 \times 3$  column and antidiagonal sum not equal to 0 3 5 6 or 7.”. It has offset 1 and begins

785, 637, 799, 916, 1305, 1894, 2310, 3280, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical:  $a(n) = 4a(n-3) - 4a(n-6) + a(n-9)$  for  $n > 14$ .

As of the “Last modified” line on the live entry (Dec 03 2018 11:52:48, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

## 2 Three consecutive lines

Write an admissible array as a sequence of columns  $u_1, u_2, \dots$ , each an element of  $\{0, \dots, 3\}^7$ . A  $3 \times 3$  subblock occupies three consecutive columns and three consecutive rows, so with  $r, s, t$  three consecutive columns the blocks it contributes are

$$g = \begin{pmatrix} r_i & s_i & t_i \\ r_{i+1} & s_{i+1} & t_{i+1} \\ r_{i+2} & s_{i+2} & t_{i+2} \end{pmatrix}$$

for each admissible index, and the entry’s condition is the requirement that

every row, diagonal sum  $\in \{0, 3, 5, 6, 7\}$  and every column, antidiagonal sum  $\notin \{0, 3, 5, 6, 7\}$

holds for every such  $g$ . This involves  $r$ ,  $s$  and  $t$  only: it is a condition on three consecutive columns and on nothing else.

Take as vertices of a digraph  $G$  the ordered PAIRS  $(r, s)$  of columns, of which there are  $(3 + 1)^{2 \cdot 7} = 268435456$ , and put an edge  $(r, s) \rightarrow (s, t)$  exactly when the condition holds for the triple  $(r, s, t)$  at every admissible index. An array with  $L$  columns is then a walk of length  $L - 2$  in  $G$ : its first two columns name the starting vertex and each further column is one step. Most of those pairs are indistinguishable, and exactly so. For a window  $j$ , whether a column  $t$  is admissible there depends on  $r$  and  $s$  only through their own  $j$ -th windows, so collect for each  $j$  the set of triples  $(t_j, t_{j+1}, t_{j+2})$  the pair allows and call that tuple  $C(r, s)$ . The columns that may follow  $(r, s)$  are exactly those consistent with every  $C_j$ , and the successor  $(s, t)$  carries  $C(s, t)$ , which does not mention  $r$ . Hence two pairs  $(r, s)$  and  $(r', s)$  with the same  $C$  have the same outgoing columns and the same successors: they may be identified, and the vertex is  $(s, C)$ . Each such vertex begins an array with the multiplicity of  $r$  that produced it, which is what  $\iota$  records. That leaves  $S = 1720956$  vertices. Hence, with  $M$  the adjacency matrix of  $G$  on  $S = 1720956$  vertices and  $\iota$  the vector of starting weights,

$$a(n) = \iota^\top M^{n-1+1} \mathbf{1},$$

the exponent being fixed by the entry's own shape and checked against its published terms. In particular  $a$  is  $C$ -finite.

### 3 The proof

**Lemma 1.** *Let  $q(t) = t^9 - \sum_i c_i t^{9-i}$  be the characteristic polynomial of the conjectured recurrence. Then the residual  $a(n) - \sum_i c_i a(n - i)$  equals  $\mathbf{1}^\top M^j q(M) \mathbf{1}$  for the corresponding walk index  $j$ , so the recurrence holds from some point on if and only if  $\mathbf{1}^\top M^j q(M) \mathbf{1} = 0$  for all large  $j$ ; and by the Cayley–Hamilton theorem it is enough to examine  $j < S$ .*

*Proof.* Factor  $M^j$  out of  $M^{j+9} - \sum_i c_i M^{j+9-i}$ . For the bound,  $M^S$  is an integer combination of  $I, M, \dots, M^{S-1}$ , so the numbers  $u_j = \mathbf{1}^\top M^j q(M) \mathbf{1}$  satisfy the monic linear recurrence given by the characteristic polynomial of  $M$ ;  $S$  consecutive zeros therefore force every later one, and the last nonzero  $u_j$  pins the threshold exactly.  $\square$

**Theorem 2.**

$$a(n) = 4a(n - 3) - 4a(n - 6) + a(n - 9)$$

for every  $n > 14$ .

*Proof.* The vector  $w = q(M) \mathbf{1}$  was formed by 9 matrix–vector products in exact integer arithmetic and  $\mathbf{1}^\top M^j w$  evaluated for  $j = 0, 1, \dots$ , again exactly. Those integers vanish from the index corresponding to  $n = 14 + 1$  onwards, and the run was continued until the working vector was identically zero, which settles all larger  $j$  at once. By Lemma 1 the recurrence holds for every  $n > 14$ . The bound is exact: at  $n = 14$  the recurrence fails on the entry's own published terms, so no larger range holds.  $\square$

### 4 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 33 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix, so no rounding or sampling enters anywhere.

## References

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